

François Duquesne

Software Engineer · Belgium / Remote

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PROFILE

Software engineer with a background in scientific research, building product-grade interfaces for technical domains. My work sits at the intersection of complex data and operational decision-making — from signal intelligence and satellite operations to hydrodynamic modelling and developer tooling. I care about making technically dense information usable: clear under complexity, reliable under pressure, and honest about what the data says.

EXPERIENCE

- Sep 2022 — Present **Software Engineer** Aerospacelab
Building product-grade interfaces for intelligence analysis and satellite operations.
- Led frontend of a SIGINT platform — interactive maps and tabular views for signal intelligence.
 - Built satellite operations platform for telemetry monitoring, telecommand management, and configuration.
 - Contributed to frontend architecture across multiple intelligence and Earth observation products.
- 2019 — Jul 2022 **Full-Stack Developer** SEALAB
Built and maintained a scientific web platform for hydrodynamic modelling workflows.
- Developed full-stack features from data ingestion and backend APIs to user-facing UI.
 - Improved usability of modelling and simulation workflows for scientific users.
- 2019 **Research Support** UCLouvain — Earth and Life Institute
Extended thesis research; contributed to hydrodynamic model quality.

EDUCATION

- 2017 — 2019 **Master's in Bioengineering** UCLouvain
Major in data analysis and management — environmental modelling.
- 2014 — 2017 **Bachelor's in Bioengineering** UCLouvain
Louvain-la-Neuve, Belgium.

PUBLICATIONS

- January 2021 **A coupled ecohydrodynamic model to predict algal blooms in Lake Titicaca** Ecological Modelling
sciencedirect.com ↗

SKILLS & STACK

Frontend	Backend / APIs	Data & Viz
TypeScript · JavaScript · Vue 3 · Nuxt · React (learning) · HTML/CSS · Tailwind	Python · Django · REST APIs · SQL · PostgreSQL · Node.js	D3.js · Vue Flow · Dashboards · Geospatial / Maps · Signal data
Tools	Scientific	Languages
Git · Docker · GitLab CI/CD · Vitest · Playwright · Bash	R · MatLab · GIS (QGIS / ArcGIS) · Mathematical modelling · LaTeX	Spanish (native) · French (native) · English (fluent)